

MICHIEL VERSCHUREN

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PROFESSIONAL SUMMARY

Senior research engineer with a decade of experience building machine learning systems and data processing pipelines at scale. I've led technical investigations into novel algorithms and mentored junior engineers while maintaining hands-on involvement in system design and implementation. My background spans distributed systems, computational linguistics, and statistical analysis, with a particular focus on making complex research problems tractable for production environments.

WORK EXPERIENCE

Staff Research Engineer

Nexus Analytics

nexusanalytics.io

Amsterdam, Netherlands

March 2021 - Present

Led the design and implementation of a real-time anomaly detection platform processing 500 million events daily across multiple data centers. Established research practices and code standards for a team of six engineers, conducting weekly technical reviews and architecture discussions. Developed novel time-series forecasting models using ensemble methods that reduced false positives by 34 percent in production. Architected a feature engineering framework in Scala that reduced model training time from 8 hours to 22 minutes through distributed computation.

Senior Research Engineer

DataFlow Systems

dataflow-systems.co

Amsterdam, Netherlands

July 2017 - February 2021

Designed and deployed a distributed graph processing system handling knowledge graphs with over 2 billion nodes, enabling new product capabilities for recommendation engines. Conducted research into approximate nearest neighbor algorithms, publishing findings that informed decisions to migrate from exact search to learned index structures. Mentored four junior engineers on systems design principles and code quality, with two advancing to senior positions. Optimized critical query paths by implementing custom indexing strategies, achieving 60 percent latency reduction for customer-facing services.

Research Engineer

Cognitive Systems Lab

University of Amsterdam

Amsterdam, Netherlands

September 2014 - June 2017

Implemented deep learning models for natural language processing tasks including named entity recognition and semantic role labeling, achieving state-of-the-art results on benchmark datasets. Collaborated with PhD students and postdocs on research papers, contributing to three publications in peer-reviewed venues on neural network

architectures. Developed a benchmark framework for evaluating NLP models across diverse languages and domains, which became the group's standard evaluation toolkit. Managed GPU computing infrastructure and implemented containerization practices that reduced model deployment time from days to minutes.

EDUCATION

Master of Science in Computer Science

University of Amsterdam

Amsterdam, Netherlands

2012 - 2014

Specialization: Machine Learning and Data Mining

Bachelor of Science in Computer Science

University of Amsterdam

Amsterdam, Netherlands

2008 - 2012

TECHNICAL SKILLS

Machine Learning: Deep Learning, Neural Networks, Ensemble Methods, Statistical Modeling, Feature Engineering, Time Series Analysis

Systems: Distributed Systems, Spark, Scala, Kubernetes, Docker, Apache Flink

Data: Python, SQL, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch

Infrastructure: AWS, Database Design, Query Optimization, Monitoring and Observability

Languages: Python, Scala, Java, SQL, Bash